

LESSON 3-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Equivalent Ratios and Tables**

Maryland Standard 6.RP.A.3a • Learning target – I can build a ratio table by scaling both rows by the same factor.

REMEMBER

- **Equivalent ratios** make the same comparison. Multiply or divide **both** terms by the same number.
- A **ratio table** lists equivalent ratios. Each column is one pair.
- Adding the same number to both terms does **not** keep the ratio.

WORKED EXAMPLE

Lemonade uses 3 lemons for every 4 cups of water.
How many cups of water go with 12 lemons?

Step 1 Find the factor: $3 \times 4 = 12$ **Step 2** Multiply the other term by the same factor: $4 \times 4 = 16$ **Answer 16 cups of water****WORD BANK****ratio** – compares two amounts, like 2 : 5**equivalent** – has the same value**ratio table** – a table of equal ratios; each column is one pair**factor** – the number you multiply by**PRACTICE***show your work • label units*

1. Trail mix uses 2 cups of nuts for every 5 cups of cereal. Complete the ratio table.

Step 1 Find what 2 was multiplied by to get each top number.**Step 2** Multiply 5 by that same number.

Nuts (cups)	2	4		10	
Cereal (cups)	5		15		35

2. Fill in the missing number so the ratios are equivalent.

Step 1 Find the factor that links the two numbers you know.**Step 2** Use the same factor on the other term.

a. $3 : 4 = 15 : \underline{\hspace{1cm}}$

b. $6 : 7 = \underline{\hspace{1cm}} : 42$

c. $18 : 24 = 3 : \underline{\hspace{1cm}}$

d. $9 : 2 = 63 : \underline{\hspace{1cm}}$

3. Select all the ratios that are equivalent to 4 : 6.

Step 1 Divide 4 : 6 by 2. Is that ratio a choice?**Step 2** Multiply 2 : 3 by 2, by 3, by 4, by 5. Look for those ratios.**Step 3** For each other choice, ask: was it multiplied or added?

☐ **A** 2 : 3

☐ **B** 8 : 12

☐ **C** 6 : 8

☐ **D** 10 : 15

4. **Bracelets.** Nadia uses **4 gold** beads for every **9 purple** beads. She uses **36 purple** beads. How many **gold** beads does she use? Show a ratio table.

Step 1 Find the factor: $9 \times \underline{\hspace{1cm}} = 36$.

Step 2 Multiply 4 by the same factor.

Show your work.

Answer _____ gold beads

5. **Paint Mix.** A paint color uses 3 cups of red for every 2 cups of white. Lou uses 21 cups of red.

Step 1 Find the factor: $3 \times \underline{\hspace{1cm}} = 21$.

Step 2 Multiply 2 by the same factor.

Step 3 Add red and white to find the total.

a. How many cups of white does Lou use? _____ cups

b. How many cups of paint does Lou make in all? _____ cups

6. **Find the Error.** Mia says $5 : 8$ is equivalent to $8 : 11$ because she added 3 to each term. Is she right? Explain and give a ratio that is equivalent to $5 : 8$.

Frame Mia is _____ because adding 3 _____. A ratio equivalent to $5 : 8$ is _____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Double a family recipe out loud. Which amounts change, and by what factor?

LESSON 3-3 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**The Bridge Strategy (Scaling Ratios)**

Maryland Standard 6.RP.A.3a • Learning target – I can use the bridge strategy to scale a ratio when I cannot multiply by a whole number.

REMEMBER

- No whole number gets you to the target? **Build a bridge.**
- Divide** both terms down to a smaller equivalent ratio, then **multiply** both terms up to the target.

WORKED EXAMPLE

A drink uses 6 scoops of mix for 8 cups of water. How much water goes with 9 scoops?

Step 1 $6 \times ? = 9$ is not a whole number. Bridge down:
 $6 : 8 \div 2 = 3 : 4$

Step 2 Scale up: $3 : 4 \times 3 = 9 : 12$

Answer 12 cups of water

WORD BANK

bridge ratio – a smaller ratio you make first
simplify – divide both numbers by the same number

target – the number you need to reach

PRACTICE

show your work • label units

1. A machine fills 15 bottles every 6 minutes. Use the bridge column to find the bottles for 10 minutes.

Step 1 $6 \div 3 = 2$. Divide 15 by 3 too.

Step 2 Then multiply the 2-minute column by 5.

Minutes	6	2	10
Bottles	15		

2. Use a bridge to find each missing number.

a. $4 : 6 = 10 : \underline{\hspace{2cm}}$

b. $9 : 12 = 6 : \underline{\hspace{2cm}}$

3. **Map Scale.** On a map, **4 cm** stand for **30 km**. How many **km** do **10 cm** stand for?

Step 1 Divide both numbers by 2 to make the bridge.

Step 2 Multiply both bridge numbers by 5.

a. Bridge ratio:

b. Kilometers for 10 cm: km

4. **Bread.** A recipe uses 10 cups of flour for 4 loaves of bread.

Step 1 Divide 10 and 4 by 2.

Step 2 Multiply the bridge by 3 to reach 6 loaves.

a. Bridge ratio:

b. Cups of flour for 6 loaves: cups

- 5. Find the Error.** A glaze uses 6 spoons honey to 10 spoons water. For 15 spoons honey, Ana added 9 to both terms: $15 : 19$. Explain her mistake and fix it.

Frame Ana's mistake is _____. The correct amount of water is _____ because _____.

TEST PRACTICE

choose the best answer

- 6.** The ratio of cups of rice to cups of water is $12 : 8$. How many cups of water go with 15 cups of rice?
- A** 10 **B** 11 **C** 12 **D** 22.5
- 7.** Which steps scale $6 : 9$ to $10 : \underline{\hspace{1cm}}$ using the bridge strategy?
- A** $\div 3$, then $\times 5$ **B** $+ 4$ **C** $\times 2$, then $- 2$ **D** $\div 6$, then $\times 9$

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Explain to someone at home why you divide first when you cannot multiply straight to the number you need.